

PSP AWS ReadWrite tip-toe steps

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Pre-requisites

- Verify latest veridata compare run to make sure there are NO OOS rows for IBOB + Monolith
 - In case of OOS, follow steps as per belowexample

```
1. log in to db as GGT user

2. run 15 days veridata compare check OOS rows in veridata for table and OOS rows and columns

3. run below update on target(12c) database as GGT user. Change table name as per OOS report.
update pspadm.PSP_ADDRESS set (ADDRESS_LINE2)=(select ADDRESS_LINE2 from pspadm.PSP_ADDRESS@to_source
where ADDRESS_SEQ='bd8ae74e-d3a4-4fb9-a2c4-eb0cf9c2c7b6' and REALM_ID='-1')
where ADDRESS_SEQ='bd8ae74e-d3a4-4fb9-a2c4-eb0cf9c2c7b6' and REALM_ID='-1';
```

- AWS & IHP - Ensure Golden gate replication is working between two

1. IHP - monolith

GGSCI (pprdpspdb800.ie.intuit.net) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPQPD4	00:00:03	00:00:05
EXTRACT	RUNNING	PPSPPP01	00:00:00	00:00:02
REPLICAT	RUNNING	RPSPPP01	00:00:00	00:00:07

AWS - monolith:

GGSCI (ggpittpp001.sbg-ppsp-prod.a.intuit.com) 1> nfo all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPPP01	00:00:03	00:00:06
EXTRACT	RUNNING	PPSPQPD4	00:00:00	00:00:09
REPLICAT	RUNNING	RPSPQPD4	00:00:00	00:00:07

IHP - IBOB

GGSCI (pprdpspdb803) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIBOBWU1	00:00:04	00:00:01
EXTRACT	RUNNING	PIBPP001	00:00:00	00:00:02
REPLICAT	RUNNING	RIBPP001	00:00:00	00:00:03

AWS - IBOB:

GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIBPP001	00:00:04	00:00:03
EXTRACT	RUNNING	PIBQPD03	00:00:00	00:00:03
REPLICAT	RUNNING	RIBQPD03	00:00:08	00:00:02

2. Check lag via heartbeat on Monolith and IBOB

login to 12c (pprdpspdb901.ie.intuit.net)

Monolith :

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;
```

IBOB :

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from ibobadm.gg_heartbeat;
```

3. Verify critical table data sanity

Monolith :

From 07/01 to 08/31 - select count(1) from pspadm.psp_financial_transaction partition(FINANCIAL_TXN_MG42019);

IBOB :

From 07/01 to 07/31 - select count(1) from ibobadm.PSP_SOURCE_SYSTEM_TRANSMISSION partition(SRCSYSTRNS_M072019);
From 08/01 to 08/31 - select count(1) from ibobadm.PSP_SOURCE_SYSTEM_TRANSMISSION partition(SRCSYSTRNS_M082019);

- AWS Monolith - Ensure all user/app accounts are locked except below ones :
 - PSPREAD_AWS, PSPBOADMIN, GGS, GGT, RDSADMIN, RDS_WAVEFRONT, SBG_WAVEFRONT
- AWS IBOB - Ensure all user/app accounts are locked except below ones :

- IBOB_PSPREAD_AWS, GGS, GGT, RDSADMIN, RDS_WAVEFRONT, SBG_WAVEFRONT
- AWS - Test and ensure cloud watch page alerts are working
- Review IOPS and Instance class settings for monolith and IBOB RDS.

Tip-toe steps

- stop service.

```
Monolith:
Login to pprdpspdb900.ie.intuit.net and stop prod service
SQL> exec dbms_service.stop_service('pspprod');

PL/SQL procedure successfully completed.

IBOB:
login to pprdpspdb903.ie.intuit.net
exec dbms_service.stop_service('ibobprod');

-----IF READ ONLY traffic will be served from IHP, then below is not required. This is not required for Tip-
toe #4
login to pprdpspdb800.ie.intuit.net and stop DR service
SQL> exec dbms_service.stop_service('pspdr');
PL/SQL procedure successfully completed.

login to pprdpspdb803.ie.intuit.net and stop DR service
SQL> exec dbms_service.stop_service('ibobdr');
PL/SQL procedure successfully completed.
```

- Lock application user accounts and keep few readonly user unlocked on IHP database.

```

1.
Monolith :
alter user PSPADM account lock;
alter user PSPAPP account lock;

/* This is not required for Tip-toe #4
alter user INTU_IQUATE account lock;
alter user OIMUSER account lock;
alter user PSPREAD account lock;
alter user PSPLOG account lock;
alter user PSPREPORTS account lock;
alter user APPBUGFIX account lock;
alter user PSPBOADMIN account lock;
*/

IBOB :
alter user IBOBADM account lock;
alter user IBOB_PROD_PSPAPP account lock;
alter user IBOB_PROD_PSPREAD account lock;

/* Below is not required for tip-toe #2, tip-toe #3
-- Unlock below ONLY IF locked in AWS
< get list of users from PD as mentioned in mail thread>

/* for reference
alter user ARCADM account unlock;
alter user DIS account unlock;
alter user EBIETL account unlock;
alter user IACETL account unlock;
alter user IAC_ETL_USER account unlock;
alter user ICADMIN account unlock;
alter user IDENTITYIQ account unlock;
alter user INTU_IQUATE account unlock;
alter user OIMUSER account unlock;
alter user PSPBOADMIN account unlock;
alter user PSPFLUXADMIN08 account unlock;
alter user PSPREAD account unlock;
alter user PSPREPORTS account unlock;
*/
*/

2.
Monolith:
Kill existing connections (if still exist)
select 'alter system kill session '||''''||sid||','||serial#||''''||' immediate;' from gv$session where
username in ('PSPADM','PSPAPP') ;

IBOB:
Kill existing connections (if still exist)
select 'alter system kill session '||''''||sid||','||serial#||''''||' immediate;' from gv$session where
username in ('IBOBADM','IBOB_PROD_PSPAPP') ;

```

- Verify account_status is LOCKED for all below users

```

Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

/* This is not required for Tip-toe #4
select username,account_status from dba_users where username in
('PSPADM','PSPAPP','INTU_IQUATE','OIMUSER','PSPREAD','PSPLOG','PSPREPORTS','APPBUGFIX','PSPBOADMIN');
*/

IBOB:
select username,account_status from dba_users where username like '%IBOB%';

/* Later For reference
select username,account_status from dba_users where username like '%PSP%' or username in
('APPBUGFIX','ARCADM','ICADMIN','EBIETL');

USERNAME                                ACCOUNT_STATUS
-----                                -
ARCADM                                LOCKED
PSPHIST_QBOE                          LOCKED
APPBUGFIX                             LOCKED
PSPLOG                                LOCKED
PSPFLUXADM                            LOCKED
PSPFLUXADMIN08                        LOCKED
PSPADM                                LOCKED
PSPTMP                                LOCKED
PSPBOADMIN                            LOCKED
PSPAPP                                LOCKED
PSPREPORTS                            LOCKED
EBIETL                                LOCKED
PSPREAD                               LOCKED
ICADMIN                               LOCKED

14 rows selected.
*/

```

- Verify GG replication

1. IHP - monolith

GGSCI (pprdpspdb800.ie.intuit.net) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPQPD4	00:00:03	00:00:05
EXTRACT	RUNNING	PPSPPP01	00:00:00	00:00:02
REPLICAT	RUNNING	RSPPPP01	00:00:00	00:00:07

AWS - monolith:

GGSCI (ggpittpp001.sbg-ppsp-prod.a.intuit.com) 1> nfo all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPPP01	00:00:03	00:00:06
EXTRACT	RUNNING	PPSPQPD4	00:00:00	00:00:09
REPLICAT	RUNNING	RPSPQPD4	00:00:00	00:00:07

IHP - IB0B

GGSCI (pprdpspdb803) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIB0BUW1	00:00:04	00:00:01
EXTRACT	RUNNING	PIBPP001	00:00:00	00:00:02
REPLICAT	RUNNING	RIBPP001	00:00:00	00:00:03

AWS - IB0B:

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIBPP001	00:00:04	00:00:03
EXTRACT	RUNNING	PIBQPD03	00:00:00	00:00:03
REPLICAT	RUNNING	RIBQPD03	00:00:08	00:00:02

- Wait for 5 mins

1. Check lag via heartbeat on Monolith and IB0B

Monolith :

```
login to 12c (pprdpspdb901.ie.intuit.net)
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;
```

IB0B :

```
login to pprdpspdb903.ie.intuit.net
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from ibobadm.gg_heartbeat;
```

- restart extract EPSPQPD4 on IHP
- check and verify sequence between IHP and AWS

```

set pagesize 1000
set numwidth 30

col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32

select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order
by SEQUENCE_OWNER, SEQUENCE_NAME;

```

- Unlock application user accounts on AWS

```

1.
Monolith :
alter user PSPADM account unlock;
alter user PSPAPP account unlock;

/* This is not required for Tip-toe #4
alter user INTU_IQUATE account unlock;
alter user OIMUSER account unlock;
alter user PSPREAD account unlock;
alter user PSPLOG account unlock;
alter user PSPREPORTS account unlock;
alter user APPBUGFIX account unlock;
alter user PSPBOADMIN account unlock;
*/

IBOB:
alter user ibobadm account unlock;
alter user IBOB_PROD_PSPAPP account unlock;
alter user IBOB_PROD_PSPREAD account unlock;

/* Below is not required for tip-toe #2, tip-toe #3
-- Unlock below ONLY IF locked in IHP
< get list of users from PD as mentioned in mail thread>

/* for reference
alter user ARCADM account unlock;
alter user DIS account unlock;
alter user EBIETL account unlock;
alter user IACETL account unlock;
alter user IAC_ETL_USER account unlock;
alter user ICADMIN account unlock;
alter user IDENTITYIQ account unlock;
alter user INTU_IQUATE account unlock;
alter user OIMUSER account unlock;
alter user PSPBOADMIN account unlock;
alter user PSPFLUXADMIN08 account unlock;
alter user PSPREAD account unlock;
alter user PSPREPORTS account unlock;
*/
*/

```

- verify accounts are unlocked

```

Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

/* This is not required for Tip-toe #4
select username,account_status from dba_users where username in
('PSPADM','PSPAPP','INTU_IQUATE','OIMUSER','PSPREAD','PSPLOG','PSPREPORTS','APPPBUGFIX','PSPBOADMIN');
*/

IBOB:
select username,account_status from dba_users where username like '%IBOB%';

/* Later For reference
select username,account_status from dba_users where username like '%PSP%' or username in
('APPPBUGFIX','ARCADM','ICADMIN','EBIETL');

```

USERNAME	ACCOUNT_STATUS
ARCADM	OPEN
PSPHIST_QBOE	OPEN
APPPBUGFIX	OPEN
PSPLOG	OPEN
PSPFLUXADM	OPEN
PSPFLUXADMIN08	OPEN
PSPADM	OPEN
PSPTMP	OPEN
PSPBOADMIN	OPEN
PSPAPP	OPEN
PSPREPORTS	OPEN
EBIETL	OPEN
PSPREAD	OPEN
ICADMIN	OPEN

```

14 rows selected.
*/

```

- Verify sequence replication between AWS and IHP (one time step)

```

set pagesize 1000
set numwidth 30

col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32

select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order
by SEQUENCE_OWNER, SEQUENCE_NAME;

```

Post tip-toe steps

- Verify GG sync and ensure no lag between AWS and IHP
- On AWS

```
stats EPSPPP01
```

Rollback

- stop application and check for connections. Kill if any

Monolith:

Kill existing connections (if still exist)

```
select 'exec rdsadmin.rdsadmin_util.kill('||s.sid||','||s.serial#||','||''IMMEDIATE''||');' from gv$session  
s where username in ('PSPADM','PSPAPP') ;
```

IBOB:

Kill existing connections (if still exist)

```
select 'exec rdsadmin.rdsadmin_util.kill('||s.sid||','||s.serial#||','||''IMMEDIATE''||');' from gv$session  
s where username in ('IBOBADM','IBOB_PROD_PSPAPP') ;
```

- Lock application user accounts on AWS database.

1.

Monolith:

```
alter user PSPADM account lock;
```

```
alter user PSPAPP account lock;
```

/* This is not required for Tip-toe #4

```
alter user INTU_IQUATE account lock;
```

```
alter user OIMUSER account lock;
```

```
alter user PSPREAD account lock;
```

```
alter user PSPLOG account lock;
```

```
alter user PSPREPORTS account lock;
```

```
alter user APPBUGFIX account lock;
```

```
alter user PSPBOADMIN account lock;
```

*/

IBOB:

```
alter user IBOBADM account lock;
```

```
alter user IBOB_PROD_PSPREAD account lock;
```

```
alter user IBOB_PROD_PSPAPP account lock;
```

- verify accounts are locked

```

1.
Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

/* This is not required for Tip-toe #4
select username,account_status from dba_users where username in
('PSPADM','PSPAPP','INTU_IQUATE','OIMUSER','PSPREAD','PSPLOG','PSPREPORTS','APPBUGFIX','PSPBOADMIN');
*/

IBOB
select username,account_status from dba_users where username like '%IBOB%';

/* For reference
Verify account_status is LOCKED for all below users
select username,account_status from dba_users where username like '%PSP%' or username in
('APPBUGFIX','ARCADM','ICADMIN','EBIETL');

```

USERNAME	ACCOUNT_STATUS
ARCADM	LOCKED
PSPHIST_QBOE	LOCKED
APPBUGFIX	LOCKED
PSPLOG	LOCKED
PSPFLUXADM	LOCKED
PSPFLUXADMIN08	LOCKED
PSPADM	LOCKED
PSPTMP	LOCKED
PSPBOADMIN	LOCKED
PSPAPP	LOCKED
PSPREPORTS	LOCKED
EBIETL	LOCKED
PSPREAD	LOCKED
ICADMIN	LOCKED

```

14 rows selected.
*/

```

- Verify GG replication

1. IHP - monolith

GGSCI (pprdpspdb800.ie.intuit.net) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPQPD4	00:00:03	00:00:05
EXTRACT	RUNNING	PPSPPP01	00:00:00	00:00:02
REPLICAT	RUNNING	RSPPPP01	00:00:00	00:00:07

AWS - monolith:

GGSCI (ggpittpp001.sbg-ppsp-prod.a.intuit.com) 1> nfo all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPPP01	00:00:03	00:00:06
EXTRACT	RUNNING	PPSPQPD4	00:00:00	00:00:09
REPLICAT	RUNNING	RPSPQPD4	00:00:00	00:00:07

IHP - IB0B

GGSCI (pprdpspdb803) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIB0BUW1	00:00:04	00:00:01
EXTRACT	RUNNING	PIBPP001	00:00:00	00:00:02
REPLICAT	RUNNING	RIBPP001	00:00:00	00:00:03

AWS - IB0B:

GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIBPP001	00:00:04	00:00:03
EXTRACT	RUNNING	PIBQPD03	00:00:00	00:00:03
REPLICAT	RUNNING	RIBQPD03	00:00:08	00:00:02

- Wait for 5 mins

1. Check lag via heartbeat on Monolith and IB0B

login to 12c (pprdpspdb901.ie.intuit.net)

Monolith :

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;
```

IB0B :

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from ibobadm.gg_heartbeat;
```

- restart extract EPSPPP01 on AWS
- check and verify sequence between AWS and IHP

```

set pagesize 1000
set numwidth 30

col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32

select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order
by SEQUENCE_OWNER, SEQUENCE_NAME;

```

- start service.

```

1.
Login to pprdspdb900.ie.intuit.net and start prod service
SQL> exec dbms_service.start_service('pspprod');

PL/SQL procedure successfully completed.

IBOB:
exec dbms_service.start_service('ibobprod');

-----Not relevant for Tip-toe #4
login to pprdspdb800.ie.intuit.net and stop DR service
SQL> exec dbms_service.start_service('pspdr');
  PL/SQL procedure successfully completed.

SQL> exec dbms_service.start_service('ibobdr');
  PL/SQL procedure successfully completed.
-----Not relevant for Tip-toe #2

```

- Unlock application user accounts on IHP

```

1.
Monolith :
alter user PSPADM account unlock;
alter user PSPAPP account unlock;

/* This is not required for Tip-toe #4
alter user INTU_IQUATE account unlock;
alter user OIMUSER account unlock;
alter user PSPREAD account unlock;
alter user PSPLOG account unlock;
alter user PSPREPORTS account unlock;
alter user APPBUGFIX account unlock;
alter user PSPBOADMIN account unlock;
*/

IBOB:
alter user ibobadm account unlock;
alter user IBOB_PROD_PSPAPP account unlock;
alter user IBOB_PROD_PSPREAD account unlock;

```

- verify accounts are unlocked

```

Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

/* This is not required for Tip-toe #4
select username,account_status from dba_users where username in
('PSPADM','PSPAPP','INTU_IQUATE','OIMUSER','PSPREAD','PSPLOG','PSPREPORTS','APPBUGFIX','PSPBOADMIN');
*/

IBOB
select username,account_status from dba_users where username like '%IBOB%';

```

- For reference ONLY

```

/* For reference. Not required for tip toe #2, tip toe#3, tip toe#4
-- Unlock below ONLY IF locked in IHP
alter user PSPBOADMIN account unlock;
alter user PSPFLUXADM account unlock;
alter user PSPLOG account unlock;
alter user PSPFLUXADMIN08 account unlock;
alter user PSPLOG account unlock;
alter user PSPREAD account unlock;
alter user PSPTEMP account unlock;
alter user APPBUGFIX account unlock;
alter user ARCADM account unlock;
alter user EBIETL account unlock;
alter user ICADMIN account unlock;

2. verify
select username,account_status from dba_users where username like '%PSP%' or username in
('APPBUGFIX','ARCADM','ICADMIN','EBIETL');

```

USERNAME	ACCOUNT_STATUS
ARCADM	OPEN
PSPHIST_QBOE	OPEN
APPBUGFIX	OPEN
PSPLOG	OPEN
PSPFLUXADM	OPEN
PSPFLUXADMIN08	OPEN
PSPADM	OPEN
PSPTEMP	OPEN
PSPBOADMIN	OPEN
PSPAPP	OPEN
PSPREPORTS	OPEN
EBIETL	OPEN
PSPREAD	OPEN
ICADMIN	OPEN

```

14 rows selected.

```

Post rollback steps

- Verify GG sync and ensure no lag between AWS and IHP
- On IHP

```
stats EPSPQPD4
```

Performance tuning

Copy SQL Plan - IHP to AWS

- from cursor cache

```
oracle@pprdpspdb900.ie.intuit.net:/home/oracle/kpopat->./create_spb_aws.sh 627bwsf9sj3g4 803259958 PSP PSP-UW2

Copying plan for sql_id 627bwsf9sj3g4 from cluster# PSP to cluster# PSP-UW2 for plan hash value 803259958

Source DBL:SPM_PSP_LAS
Target DBL:SPM_PSP_UW2
Exporting sql plan for sqlid [627bwsf9sj3g4]
SQL plan export complated!
Importing sql plan for sqlid [627bwsf9sj3g4]
SQL plan import complated!
Export/Import of sql plan completed for sqlid [627bwsf9sj3g4] plan hash [803259958]

PL/SQL procedure successfully completed.
```

- from AWR repository

```
oracle@pprdpspdb900.ie.intuit.net:/home/oracle/kpopat->./create_spb_aws_awr.sh 1wbsa04w60gzt 3307958781 PSP PSP-UW2 4085 4086

Copying plan for sql_id 1wbsa04w60gzt from cluster# PSP to cluster# PSP-UW2 for plan hash value 3307958781

Source DBL:SPM_PSP_LAS
Target DBL:SPM_PSP_UW2
Exporting sql plan for sqlid [1wbsa04w60gzt]
SQL plan export complated!
Importing sql plan for sqlid [1wbsa04w60gzt]
SQL plan import complated!
Export/Import of sql plan completed for sqlid [1wbsa04w60gzt] plan hash [3307958781]

PL/SQL procedure successfully completed.
```